Biconnected Graph Triangulation

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Abstract

First we will show the higher level pseudocode of our algorithm for enumerating all possible triangulations of a biconnected plane graph. Then we will illustrate the steps of the algorithm with an example graph.

FindAllTriangulations

Algorithm FindAllTriangulations

procedure $\textsc{FindAllTriangulations}(G, k)$
    Label faces as $F_1, F_2, \ldots, F_k$ arbitrarily
    Initialize global chord set S
    $T \leftarrow \emptyset$ ▷ Empty partial triangulation
    $\textsc{FindAllTriangulationsCycle}(F, T, 0)$
end procedure

FindSafeRootTriangulation

Algorithm FindSafeRootTriangulation

```
procedure FindSafeRootTriangulation(F, index)
    startIndex ← n − 1
    endIndex ← 0
    while startIndex > endIndex + 1 do
        if chord (v_startIndex, v_endIndex) exists in S then
            startIndex ← startIndex − 1
        else
            endIndex ← endIndex + 1
        end if
    end while
    v_r ← v_endIndex                    ▷ Root vertex for safe triangulation
    Build fan triangulation T_r from v_r as root vertex
    return T_r
end procedure
```

FindAllTriangulationsCycle

Algorithm FindAllTriangulationsCycle

procedure FindAllTriangulationsCycle(F, T, i)
    $T_i \leftarrow$ FindSafeRootTriangulation(F, i) ▷ $\mathcal{O}(n)$
    if F_i is the last face **then** ▷ All faces triangulated
        output $(T \cup T_i)$
    else
        FindAllTriangulationsCycle($F, T \cup T_i, i + 1$)
    end if
    for $j = n - 1$ **down to** 3 **do**
        FindAllChildTriangulationsCycle($T_i(v_r, v_j), T, i$)
    end for
end procedure

FindAllChildTriangulationsCycle

Algorithm FindAllChildTriangulationsCycle

procedure $\textsc{FindAllChildTriangulationsCycle}(T_i, T, i)$
  if F_i is the last face **then** ▷ All faces triangulated
    output $(T \cup T_i)$
  else
    $\textsc{FindAllTriangulationsCycle}(F, T \cup T_i, i+1)$
  end if
  Let $(v_b, v_{b'})$ be the leftmost generating chord of T_i
  for j from b to $n-1$ **do**
    if (v_r, v_j) is a chord of T_i **or** $\mathit{oppositePair}(v_r, v_j) \notin S$ **then**
      $\textsc{FindAllChildTriangulationsCycle}(T_i(v_r, v_j), T, i)$
    end if
  end for
end procedure

A biconnected plane graph with three faces

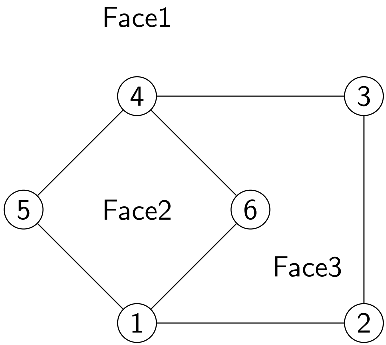
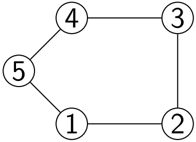
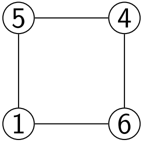


Figure: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

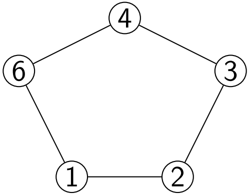
Faces of the Graph



(a) Face 1

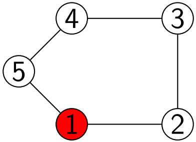


(b) Face 2

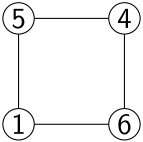


(c) Face 3

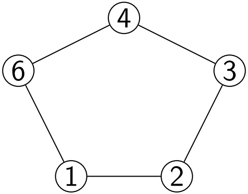
Find safe root of face 1



(a) **Face 1**



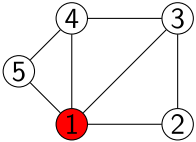
(b) Face 2



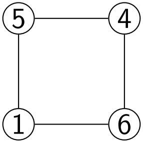
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6)

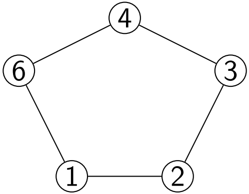
Root Triangulation of face 1



(a) **Face 1**



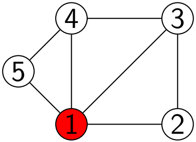
(b) Face 2



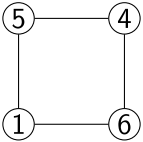
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

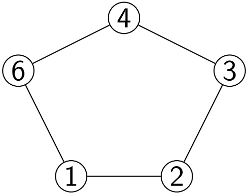
Now we move to face 2



(a) Face 1



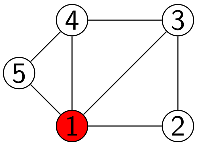
(b) **Face 2**



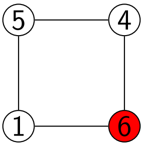
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

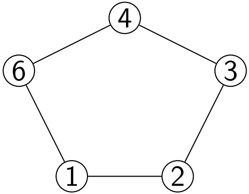
Find safe root of face 2



(a) Face 1



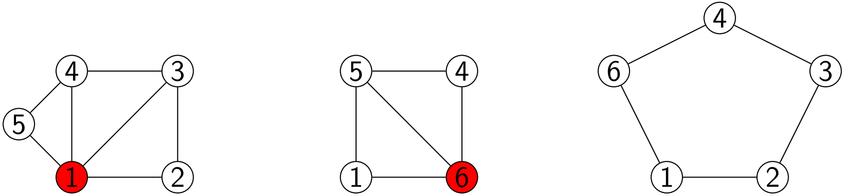
(b) **Face 2**



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

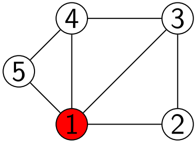
Root Triangulation of face 2



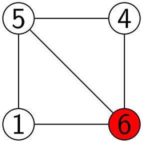
(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

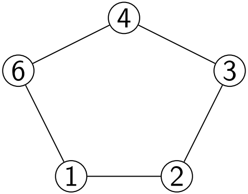
Now we move to face 3



(a) Face 1



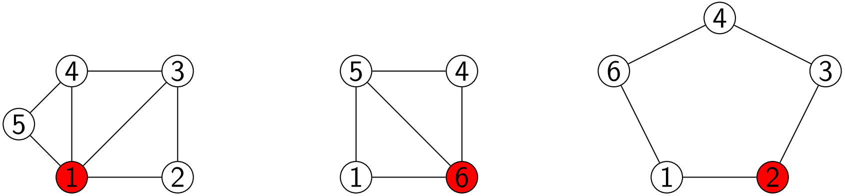
(b) Face 2



(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

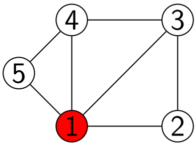
Find safe root of face 3



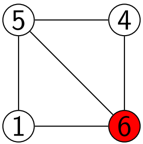
(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

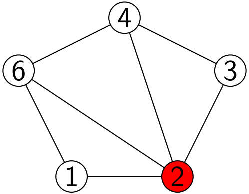
Root Triangulation of face 3



(a) Face 1



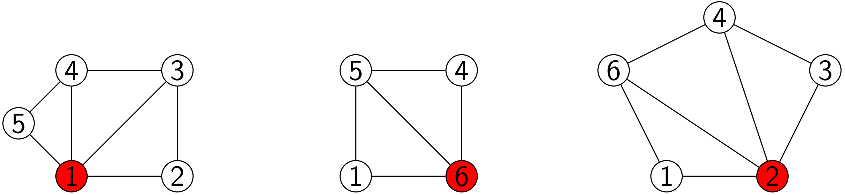
(b) Face 2



(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

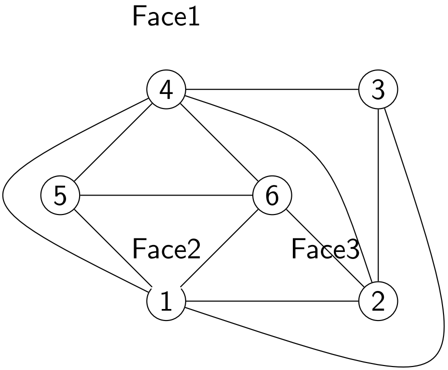
There is no other faces after face 3, so its a complete valid triangulation



(a) Face 1 (b) Face 2 (c) **Face 3**

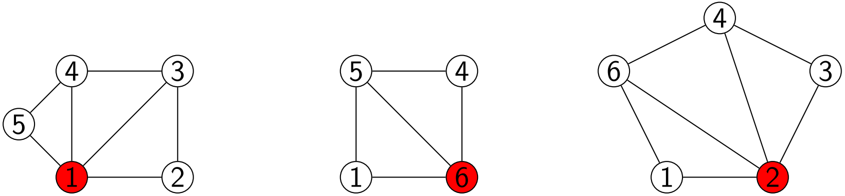
Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

So the actual triangulation looks like this:



Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

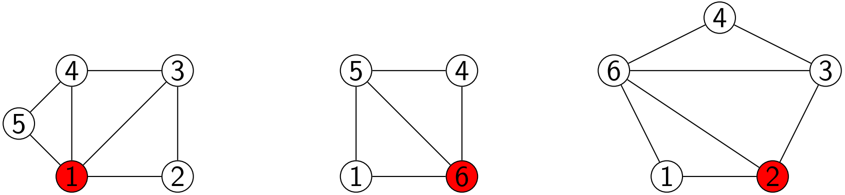
Now we will start flipping of Face 3



(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

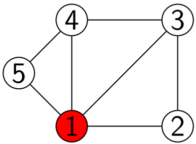
We can flip the (2,4) edge, as its opposite pair is not in the global edge set



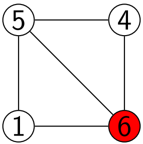
(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (3,6), (2,6)**

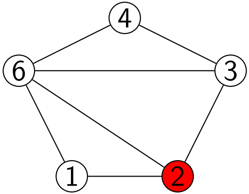
Its a new triangulation



(a) Face 1



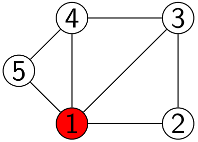
(b) Face 2



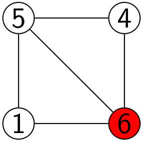
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (3,6), (2,6)**

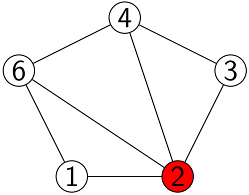
in the root also, cant flip (2,6) as (1,4) is present in the global edge set



(a) Face 1



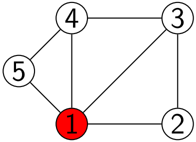
(b) Face 2



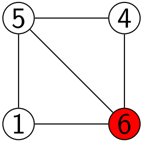
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6). (2,4), (2,6)**

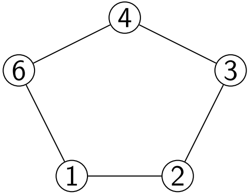
So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



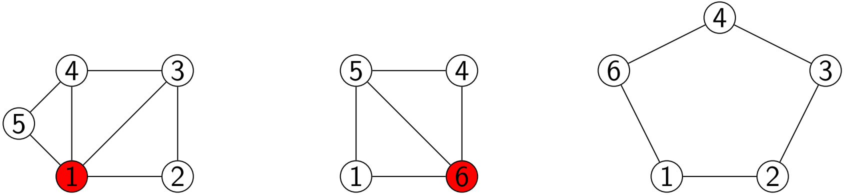
(b) **Face 2**



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

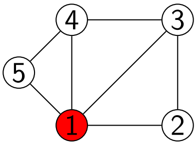
We cant flip the (5,6) edge as (1,4) is present in the global edge set



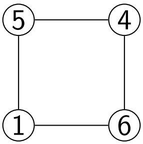
(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

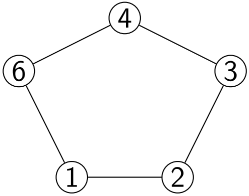
So we backtrack to face 1



(a) **Face 1**



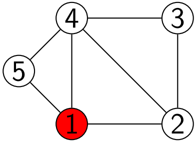
(b) Face 2



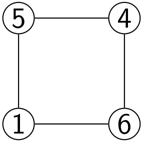
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

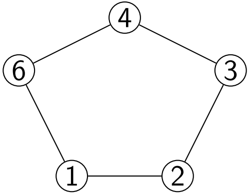
We can flip the (1,3) edge, as its opposite pair is not in the global edge set



(a) **Face 1**



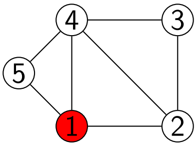
(b) Face 2



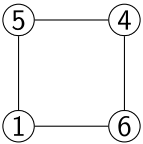
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

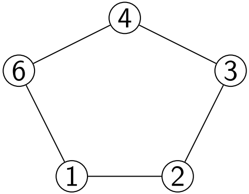
Now we move to face 2



(a) Face 1



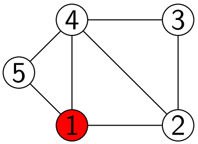
(b) **Face 2**



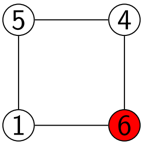
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

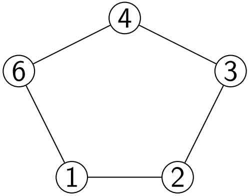
Find the safe root of face 2



(a) Face 1



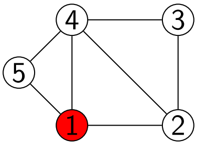
(b) **Face 2**



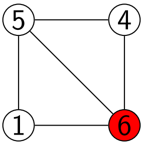
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

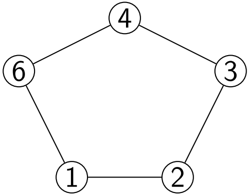
Root Triangulation of face 2



(a) Face 1



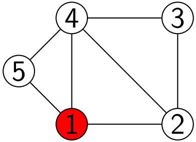
(b) **Face 2**



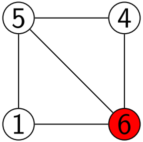
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

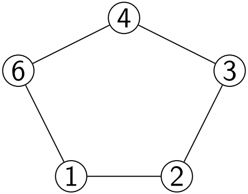
Now we move to face 3



(a) Face 1



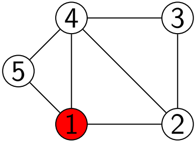
(b) Face 2



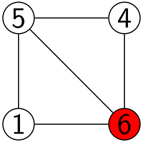
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

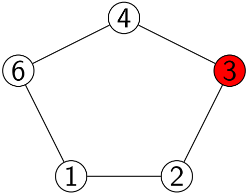
Find safe root of face 3



(a) Face 1



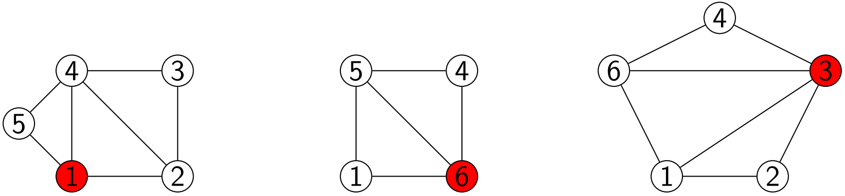
(b) Face 2



(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4), (5,6)**

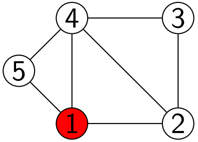
Root Triangulation of face 3



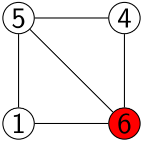
(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

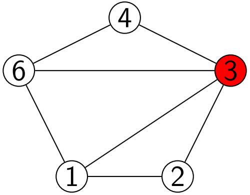
There is no face after face 3, so its a complete triangulation



(a) Face 1



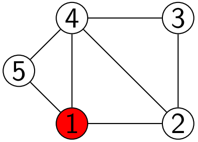
(b) Face 2



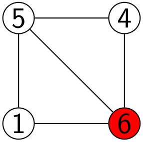
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

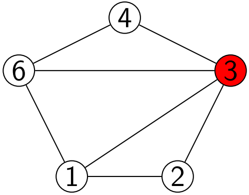
Cant flip (3,6), because opposite pair is (1,4), already exists



(a) Face 1



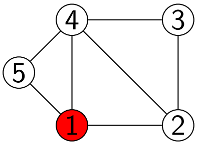
(b) Face 2



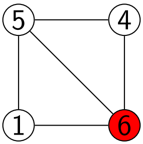
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (1,3), (3,6)**

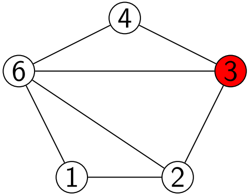
we can flip (1,3) to (2,6), its a new triangulation



(a) Face 1



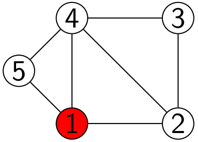
(b) Face 2



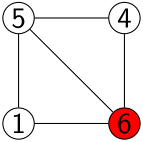
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (2,6), (3,6)**

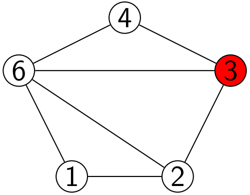
Cant flip (3,6), because opposite pair is (2,4), already exists



(a) Face 1



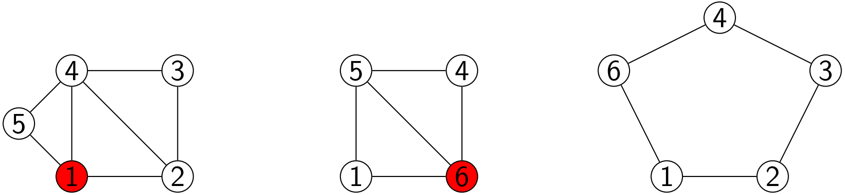
(b) Face 2



(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6), (2,6), (3,6)**

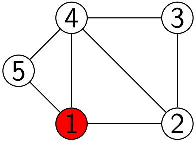
So, there is no more triangulations possible from here, so we backtrack to face 2



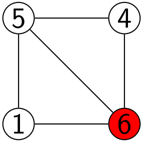
(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6)**

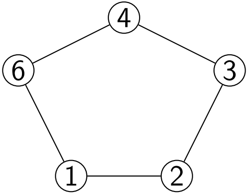
We cant flip the (5,6) edge as (1,4) is present in the global edge set



(a) Face 1



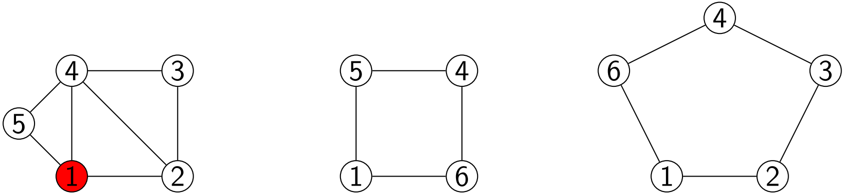
(b) **Face 2**



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2, 4), (1,4), (5,6)**

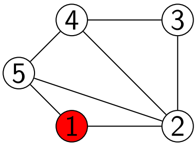
So we backtrack to face 1



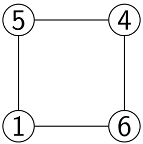
(a) **Face 1** (b) Face 2 (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (1,4)**

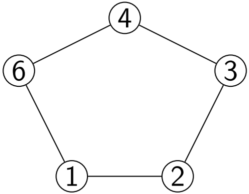
We can flip (1,4) to (2,5)



(a) **Face 1**



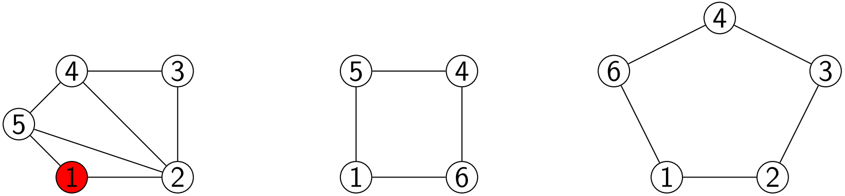
(b) Face 2



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

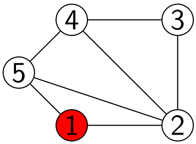
We can now move to Face 2



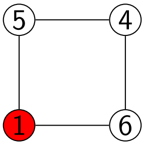
(a) Face 1 (b) **Face 2** (c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

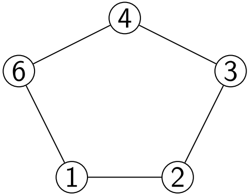
Find safe root of Face 2



(a) Face 1



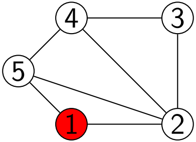
(b) **Face 2**



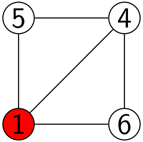
(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

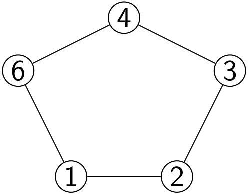
Root Triangulation of Face 2



(a) Face 1



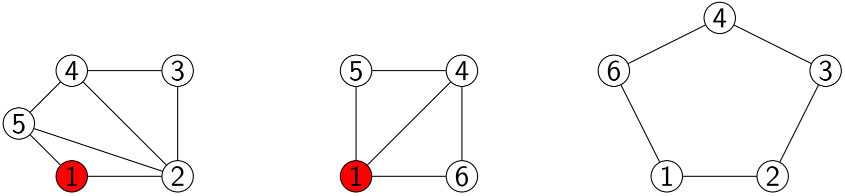
(b) **Face 2**



(c) Face 3

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

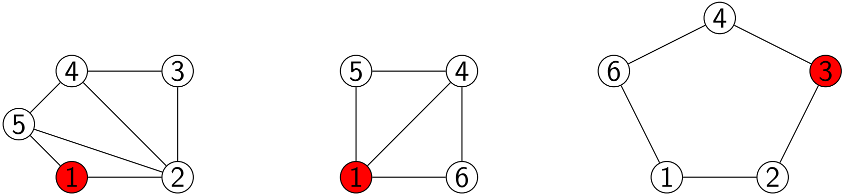
Now we move to Face 3



(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

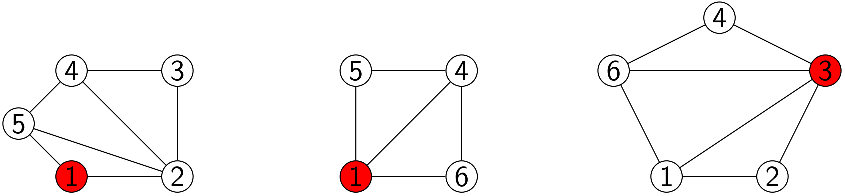
Find safe root of Face 3



(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4)**

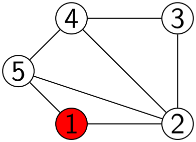
Root Triangulation of Face 3



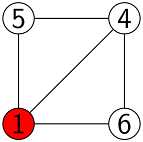
(a) Face 1 (b) Face 2 (c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (1,3)**

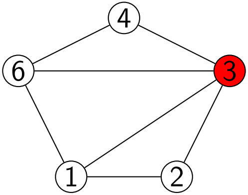
There is no more faces, so its a complete triangulation



(a) Face 1



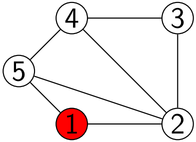
(b) Face 2



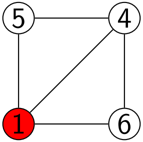
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (1,3)**

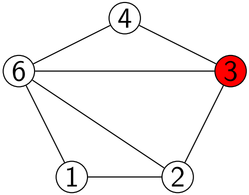
We can flip the (1,3) edge to (2,6) to get a different triangulation



(a) Face 1



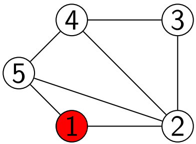
(b) Face 2



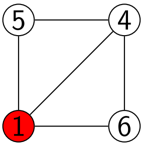
(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (2,6)**

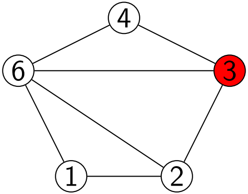
There is no more faces, so it's a complete triangulation



(a) Face 1



(b) Face 2



(c) **Face 3**

Global Edge Set:
(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5), (1,4), (3,6), (2,6)**

Continue....

We will be continuing this process until all possible triangulations are found.